

Sophia Zhang

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EDUCATION

Massachusetts Institute of Technology

Bachelor's in CS and Math

Cambridge, MA

Intended: Aug. 2025 – May 2028

Princeton International School of Mathematics and Science (PRISMS)

High school

Princeton, NJ

Sep. 2022 – June 2025

EXPERIENCE

Quantitative Research Intern

Citadel Securities

Summer 2027

Miami, FL

- Incoming Intern

AI Engineering Intern

Alpha School

June – Aug. 2026

Austin, TX

- Built apps, evaluations, and RL training setups for improving agent performance, with a focus on education and learning science applications.

Undergraduate Research Opportunities Program

Dina Katabi's Lab at MIT CSAIL

Sep. 2025 – Present

Cambridge, MA

- Working with multimodal data, mostly EEG and breathing

Trading and Technology Internship

Hudson River Trading

Jan. 2026

New York City, NY

- Worked on many projects in both software engineering and quant research, including trading Brazilian equities.

LLMs for Formal Theorem Proving

Microsoft Research Asia

June 2025 – Jan. 2026

Remote

- Helped verify math proofs in Lean and discussed findings to improve the system.

PUBLICATIONS

STAR-PólyaMath: Multi-Agent Reasoning under Persistent Meta-Strategic Supervision

arXiv 2026

- Jiaao Wu, Xian Zhang, Hanzhang Liu, **Sophia Zhang**, Fan Yang, Yinpeng Dong

Euclean: Automated Geometry Problem Formalization with Unified Verification in Lean

ICML 2026

- Linbin Tang, Jingyan You, Zilin Kang, Hanzhang Liu, **Sophia Zhang**, Zenan Li, Chenrui Cao, Liangcheng Song, Jiaao Wu, Xian Zhang, Fan Yang

PROJECTS

AI Math Olympiad Progress Prize 3 | *Python, PyTorch, vLLM, Unsloth*

Jan. 2026 – April 2026

- Worked on project with other MIT freshmen, whom were all Olympiad winners from many countries. My role was focused most on scaffolding; involved training, hidden state extraction, parameter sweeping, etc.
- Authored our comprehensive technical writeup

AWARDS

Putnam Top 200 2026

USAMO Silver 2025

USAPhO Honorable Mention 2025

Two-time Mathematical Olympiad Program (MOP) Attendee 2023, 2024

TECHNICAL SKILLS

Languages: Python, C++, C, RISC-V Assembly, Minispec, Java, Rust, HTML/CSS, JavaScript, TypeScript,

AI Research: Diffusion models, SpecDec, vLLM

Data Engineering: NumPy, SciPy, Pandas, Matplotlib, Jupyter, Automated Eval Pipelines